higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

ELECTROTECHNICS N6

7 AUGUST 2019

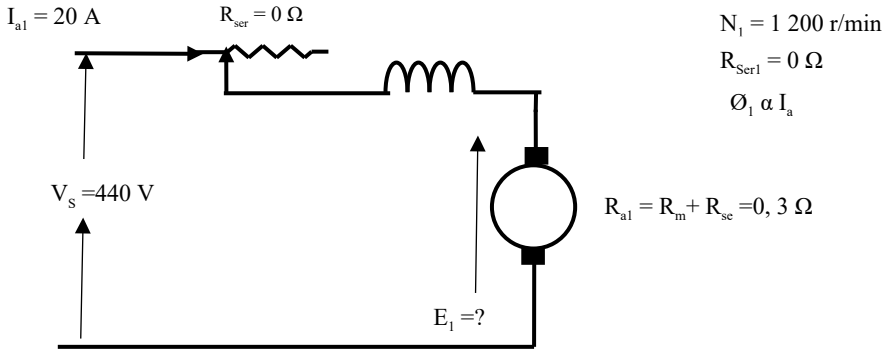
This marking guideline consists of 11 pages.

Copyright reserved

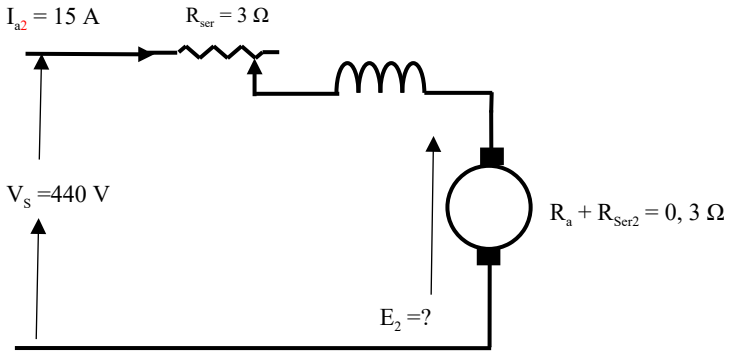
QUESTION 1: DC MACHINES

1.1 Given: A series motor

.



1.1.1



$$E_1 = V - I_{a1}R_{a1} = 440 - (20 \times 0,\ 3) = 440 - 6 = 434\ V \ \checkmark$$

$$E_2 = V - I_{a2}R_{a2} = 440 - 15(3 + 0,\ 3) = 440 - 49,5$$
$$= 390,\ 5\ V \ \checkmark$$

$$\frac{E_2}{E_1} = \frac{N_2 Ø_2}{N_1 Ø_1} \quad \therefore\ N_2 = \frac{E_2}{E_1} \times \frac{N_1 Ø_1}{Ø_2} = \frac{390,\ 5}{434} \times \frac{1\,200 \times 1}{0,\ 8}$$

= 1 349,654 r/min ✓ (6)

1.1.2 $P_1 \propto Ø_1 \cdot N_1 \cdot I_{a1}$ $P_2 \propto Ø_2 \cdot N_2 \cdot I_{a2}$

$$\frac{P_1}{P_2} = \frac{Ø_1 \cdot N_1 \cdot I_{a1}}{Ø_2 \cdot N_2 \cdot I_{a2}} = \frac{Ø_1 \times 1\,200 \times 20}{0,\ 8 \times Ø_1 \times 1349,\ 654 \times 15}$$
$$= 1,\ 482$$

(2)

1.2 Advantages:
- It is convenient and economical.
- The efficiency can be predetermined at any load.

Disadvantages:
- No account is taken of the change in iron losses from no-load to full-load.
- As the test is on no-load, it is impossible to know whether the commutation would be satisfactory at full-load and whether the temperature rise would be within the specified limit. (2 + 2) (4)

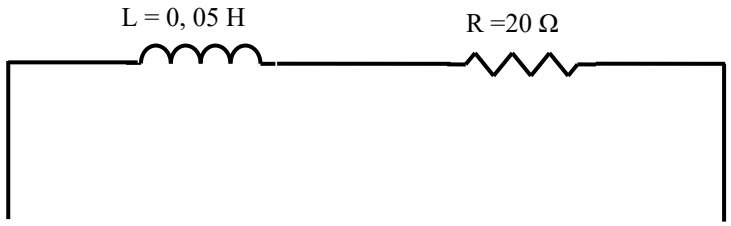
1.3 A DC machine works at maximum efficiency when the constant losses✓ and the armature copper losses are equal.✓ (2)

[14]

QUESTION 2: AC CIRCUIT THEORY

2.1
- Synchronous motors (or capacitors)
- Static capacitors
- Phase advancers (Any 2 × 1) (2)

2.2 Given: Series circuit consisting of a resistor and an inductor.
Take ω = 314 rad/s.



$e = 250 \sin \omega t + 50 \sin(3\omega t + \pi/3) + 20 \sin(5\omega t + 5\pi/6)$ volts

2.2.1 $$i = \frac{250}{25,\ 431}\sin(\omega t - 38,\ 146°) + \frac{50}{51,\ 192}\sin(3\omega t + 60° - 67°) + \frac{20}{81{,}065}\sin$$
$$= 9,\ 831\sin(\omega t - 38,\ 146°) + 0,\ 977\sin(3\omega t - 7°) + 0,\ 247\sin(5\omega t + 74,$$ (4)

2.2.2 $$I_{RMS} = \sqrt{\frac{I_{M1}^{\ 2}}{2} + \frac{I_{M3}^{\ 2}}{2} + \frac{I_{M5}^{\ 2}}{2}}$$
$$¿ \sqrt{\frac{(9,\ 831)^2}{2} + \frac{(0,\ 977)^2}{2} + \frac{(0,\ 247)^2}{2}}$$
$$¿ 6,\ 998\ A$$ (2)

2.2.3 $$V_{RMS} = \sqrt{\frac{V_{M1}^{\ 2}}{2} + \frac{V_{M3}^{\ 2}}{2} + \frac{V_{M5}^{\ 2}}{2}} = \sqrt{\frac{(250)^2}{2} + \frac{(50)^2}{2} + \frac{(20)^2}{2}}$$
$$¿ 180,\ 831\ V$$ (2)

2.2.4 $P_T = I^2 R = 48.972 \times 20 = 979,44\ W$ (2)

2.2.5 $pf = \frac{Watts}{VI} = \frac{979,44}{180,831 \times 6,998} = 0,774$ (2)

[14]

QUESTION 3: TRANSFORMERS

Given: A 50 kVA, 4 400/220 V transformer

R_1 = 3, 45 Ω X_1 = 5, 2 Ω
R_2 = 0, 009 Ω X_2 = 0, 015 Ω.

3.1 $Turns'\ ratio: \frac{T_1}{T_2} = \frac{V_1}{V_2} = \frac{4\ 400}{220} = \frac{20}{1}$

$R^1{}_2 = R_2 \times \left(\frac{T_1}{T_2}\right)^2 = 0,009 \times \left(\frac{20}{1}\right)^2 = 3,6\ \Omega$ (4)

3.2 $X^1{}_2 = X_2 \times \left(\frac{T_1}{T_2}\right)^2 = 0,015 \times \left(\frac{20}{1}\right)^2 = 6\ \Omega$ (2)

3.3 $R_{pT} = R_1 + R^1{}_2 = 3,45 + 3,6$ $X_{pT} = X_1 + X^1{}_2 = 5,2 + 6$

¿7, 05 Ω ¿11, 2 Ω (4)

3.4 **Method 1**

$Z_{pT} = R_{pT} + X_{pT} = 7,05 + j\,11,2$
$= 13,234\ 57,811° \ \Omega$

OR

Method 2

$Z_{pT} = \sqrt{R_{pT}{}^2 + X_{pT}{}^2} = \sqrt{(7,05)^2 + (11,2)^2} = 13,234\ \Omega$ (2)

3.5 $I_1 = \frac{S}{V_1} = \frac{50 \times 10^3}{4\ 400} = 11,\ 364\ \ A$

$I_2 = \frac{S}{V_2} = \frac{50 \times 10^3}{220} = 227,\ 273\ \ A$

$Cu\ \ loss = I_1^{\ 2} \cdot R_1 + \ I_2^{\ 2} \cdot R_2$

¿$\left(11{,}364^2 \times 3,\ 45\right) + \left(227,\ 273^2 \times 0{,}009\right)$√

$= 910,\ 412\ \ W$√

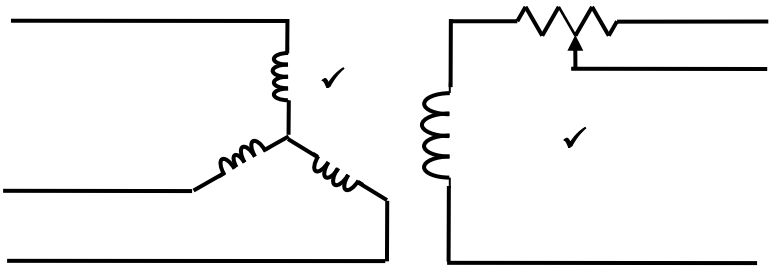
AND

$Cu = I_1^{\ 2} R_{pT} = \left(11,\ 364\right)^2 \times 7,\ 05$ √$= 910,\ 441\ \ \Omega$√ (4)

[16]

QUESTION 4: AC MACHINES – ALTERNATORS

4.1



(5)

4.2 Given: A 3-phase, 50-Hz, star-connected alternator

P = 2 000 kW X_s = 3 ohms per/phase

V_L = 11 kV pf = 0, 8 lagging

R_a = 0,25 ohms/phase

Calculating the percentage regulation.

$P = \sqrt{3} \cdot V_L \cdot I_L \cdot \cos\theta$

$I_L = \frac{P}{\sqrt{3} \cdot V_L} = \frac{2\ \ 000 \times 10^3}{\sqrt{3} \times 11\ \ 000} = 104,\ 973\ \ A$

$V_p = \frac{V_L}{\sqrt{3}} = \frac{11\ \ 000}{\sqrt{3}} = 6350,\ 853\ \ V$

$I\,R_a = 104,\ 973 \times 0{,}25 = 26{,}243\ \ 0°\ \ V$

$I\,X_S = 104{,}973 \times 3 = 314,\ 919\ \ V\ \ 90°\ \ V$

Method 1

$$E_p = V_p\theta + I R_a 0° + I X_s 90°$$
$$= 6350,\ 853\ 36,\ 87° + 26{,}243\ \ 0° + 314,\ 919\ \ 90°$$
$$= 6565,\ 04938,\ 932°\ \ V$$

$$\%\ \ Regulation = \frac{E_p - V_p}{V_p}$$

$$\text{¿} \frac{6565{,}049 - 6350,\ 853}{6350,\ 853} \times 100\%$$

$$\text{¿} 3,\ 373\%$$

OR

Method 2

$$E_p = \sqrt{\left(V_p \cdot \cos\theta + I R_a\right)^2 + \left(V_p \cdot \sin\theta + I X_s\right)^2}$$

$$\text{¿} \sqrt{(6350,\ 853 \times 0{,}8 + 26,\ 243)^2 + (6350,\ 853 \times 0,\ 6 + 314,\ 919)^2}$$

$$= 6565,\ 049\ \ V$$

$$\%\ \ Regulation = \frac{E_p - V_p}{V_p} \times 100\%$$

$$\text{¿} \frac{6565,\ 049 - 6350,\ 853}{6350,\ 853} \times 100\%$$

$$= 3,\ 373\%$$

OR

Method 3

$$E_p = V_p + \left(I R_a \cdot \cos\theta\right) + \left(I X_s \cdot \sin\theta\right)$$
$$= 6350{,}853 + (26{,}243 \times 0,\ 8) + (314,\ 919 \times 0,\ 6)$$
$$= 6560,\ 799\ \ V$$

$$\%\ \ Regulation = \frac{E_p - V_p}{V_p} \times 100\%$$

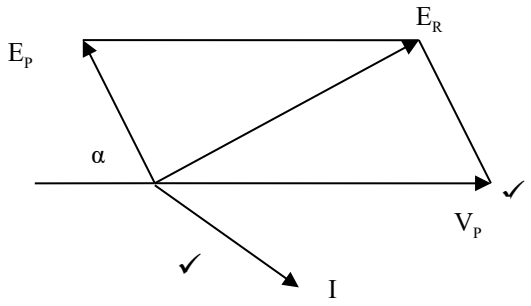
$$\text{¿} \frac{6560,\ 799 - 6350,\ 853}{6350,\ 853} \times 100\%$$

$$= 3,\ 306\%$$

(9)
[14]

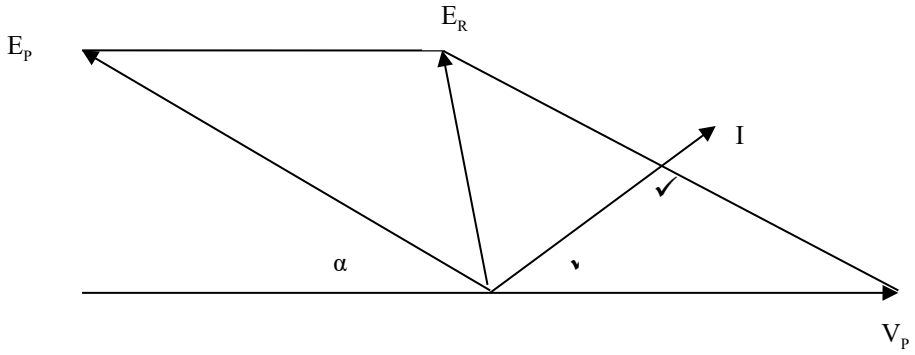
QUESTION 5: AC MACHINES – SYNCHRONOUS MOTORS

5.1 Lagging power factor



Leading power factor

Increase in excitation increases Ep✓ causes the resultant current to lead.✓



(6)

5.2. Given: a three-phase, 50 Hz, star-connected synchronous motor.

S = 5 kVA P = 3,5 kW
V_L = 400 V %Z = (4 + j40)%

5.2.1 $\cos\theta = \frac{P}{S} = \frac{3,5}{5} = 0,7$ (1)

5.2.2 $V_L = \sqrt{3} \cdot V_p$

$\therefore \quad V_p = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230,94 \ V$

$\%R = \frac{IR}{V_p}$

$IR = \%R \cdot V_p = 0,04 \times 230,\ 94$

¿9, 238 V

$\%X = \frac{IX}{V_p}$

$IX = \%X \cdot V_p = 0,4 \times 230,\ 94 = 92,376 \ V$

(3)

5.2.3 **Method 1**

$E_p = Vp - + IRa\,180 + I.Xs\text{-}90$

$¿230,94\text{-}45,573° + 9,238\,180° + 92,376\text{-}90°$

$¿299,057\text{-}59,358°\ V$

$E_L = \sqrt{3}\ .\ Ep = \sqrt{3} \times 299,057 = 517,982\ V$

OR

Method 2

$S = \sqrt{3} \cdot V_L \cdot I_L$

$I_L = \frac{S}{\sqrt{3} \cdot V_L} = \frac{5 \times 10^3}{\sqrt{3} \times 400} = 7,217\ A$

$\%R = \frac{IR}{V_p}$

$R = \frac{\%R \cdot V_p}{I} = \frac{0,04 \times 230,94}{7,217} = 1,28\ \Omega$

$\%X = \frac{IX}{V_p}$

$X = \frac{\%X \cdot V_p}{I} = \frac{0,4 \times 230,94}{7,217} = 12,8\ \Omega$

$Z_S = R + jX_S = 1,28 + j12,8 = 12,864\quad 84,289$

$E_r = I\,Z_S = 7,217\,45,573 \times\ 12,864\,84,289$

$= 92,84\,129,862°\ V$

$Ep = Er - Vp = 92,84\,129,862\quad 230,94\,0$

$= 299,060\,166,214°\ V$

$E_L = \sqrt{3} \cdot E_p = \sqrt{3} \times 299,060 = 517,987\ V$ (3)

5.2.4

Method 1		**Method 2**
$\alpha = 59,358^o - 45,573^o$	**OR**	$\alpha = 180 - 166,214^o$
$= 13,785^o$		$= 13,786^o$

(1)
[14]

QUESTION 6: AC MACHINES - INDUCTION MOTORS

6.1 Plugging is the stopping of an electric motor✓ by immediately reversing its direction until it stops.✓ (2)

6.2
- Stator
- Rotor

(2)

6.3 Given:

No of poles =6	N_2 = 950 r/min.
f = 50 Hz	$P_{stator\ loss}$ = 1 kW
P_{in} = 45 kW	$P_{fric\&wind\ loss}$ = 1,5 kW

6.3.1

$$N_1 = \frac{60 \cdot f}{p} = \frac{60 \times 50}{3} = 1\,000 \ \ r/min$$

$$S = \frac{N_1 - N_2}{N_1} \times 100\,\%$$

$$\text{¿} \frac{1\,000 - 950}{1\,000} \times 100\,\%$$

$$= 5\,\%$$

(3)

6.3.2

$$P_{in} = Stator\ \ output + Stator\ \ loss$$

$$Stator\ \ output\ \ = P_{in} - Stator\ \ loss = \ 45 - 1 = 44\ \ kW$$

$$Stator\ \ output = Rotor\ \ input = 44 \text{ kW}$$

$$Rotor\ \ input = \frac{2 \cdot \pi \cdot N_1 . T}{60}$$

$$T = \frac{60 \cdot Rotor\ \ input}{2 \cdot \pi \cdot N_1} = \frac{60 \times 44 \times 1\,000}{2 \times \pi \times 1\,000}$$

$$= 420,\ 169\ \ Nm$$

$$Rotor\ \ cu.\ \ loss = \frac{2 \cdot \pi \cdot (N_1 - N_2) T}{60}$$

$$\text{¿} \frac{2 \times \pi \times (1\,000 - 950) \times 420,\ 169}{60}$$

$$= 2,2\ \ kW$$

(4)

6.3.3

$$P_{in} = P_{out} + P_{stator} + P_{rotor} + P_{friction}$$

$$P_{out} = P_{in} - Total\ \ losses$$

$$= 45 - (1 + 1.5 + 2,2)$$

$$\text{¿} 40,3\ \ kW$$

(2)

6.3.4

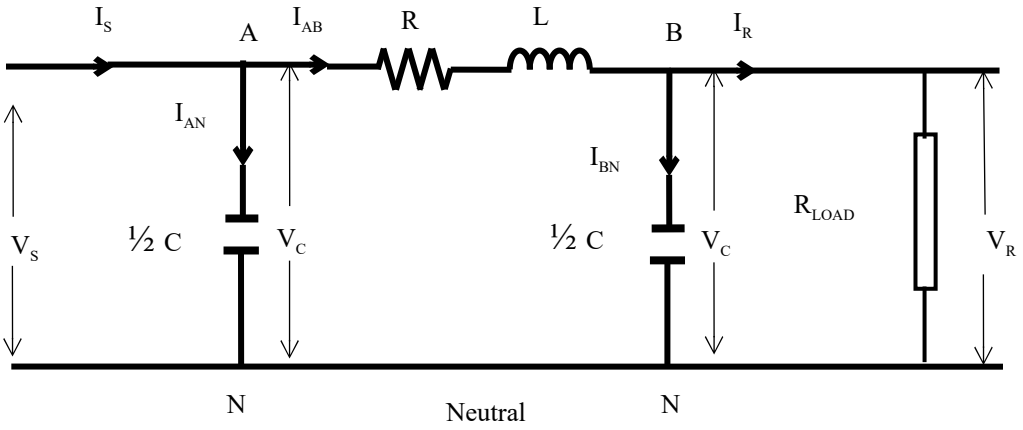
$$= \frac{P_{out}}{P_{in}} \times 100\,\% = \frac{40,3}{45} \times 100\,\% = 89,\ 556\,\%$$

(1)
[14]

QUESTION 7: GENERATION AND DISTRIBUTION OF AC

Given*:* A 3-phase transmission line

S = 75 MVA C = 0, 03 µF/km/phase
V_L = 132 kV R = 0, 28 Ω/km/phase f = 50 Hz pf = 0,8 lagging X_L = 0,63 Ω/km/phase



$X_{L/ph} = 0,\ 63 \times 100 = 63\ \Omega$
$R_{/ph} = 0,\ 28 \times 100 = 28\ \Omega$

$Z_{AB} = R + j\,X_L = 28 + j\,63 = 68,\ 942\,66{,}038°\ \Omega$

$$\frac{C}{2} = \frac{0,\ 03 \times 10^{-6} \times 100}{2} = 1{,}5\ \mu F$$

$$X_C = \frac{1}{2 \cdot \pi \cdot f \cdot \frac{C}{2}} = \frac{1}{2 \times \pi \times 50 \times 1,\ 5 \times 10^{-6}} = 2\,122,\ 066\ \Omega$$

$Z_{BN} = 0 - j\,X_C = 0 - j\,2\,122{,}066 = 2\,122,\ 066\,{-}90°\ \Omega$

$$V_R = \frac{V_L}{\sqrt{3}} = \frac{132 \times 10^3}{\sqrt{3}} = 76\,210,\ 236\,0°\ V$$

$$I_{BN} = \frac{V_R}{Z_{BN}} = \frac{76\,210,\ 236\,0°}{2\,122,\ 066\,{-}90°} = 35,\ 913\,90°\ A$$

$S = \sqrt{3} \cdot V_L \cdot I_L$

$$I_L = \frac{S}{\sqrt{3} \cdot V_L} = \frac{75 \times 10^6}{\sqrt{3} \times 132 \times 10^3} = 328,\ 04\,36,\ 87°\ A$$

$I_{AB} = I_R + I_{BN} = 328,\ 04 - 36{,}87\ + 35,\ 913\,90°$

$= 307,836 - 31,515 \quad A$

$V_{AB} = I_{AB} \cdot Z_{AB} = 307,836 - 31,515° \times 68,942\,66,038$
$= 21\,222,8334,523° \quad V$

$V_S = V_R + V_{AB} = 76\,210,2360° + 21\,222,8334,523° \quad V$
¿$94\,464,557,315° \quad V$

$V_L = \sqrt{3} \cdot V_S = \sqrt{3} \times 94\,464,55 = 163\,617,4 \quad V$

$I_{AN} = \dfrac{V_S}{Z_{AN}} = \dfrac{163\,617,47,315°}{2\,122,066\text{-}90°}$ $Z_{AN} = Z_{BN} = 2\,122,066\angle -90^{\circ}$

¿$77,10397{,}315° \quad A$

$I_S = I_{AB} + I_{AN} = 307,836 - 31,515 + 77,10397{,}315°$
$= 266,352 - 18,482° \quad A$

$Power \quad factor = \cos = \cos(18{,}482 + 7,315)$
$= \cos 25,797 = 0,9 \quad Leading$ **[14]**

TOTAL: 100

Copyright reserved